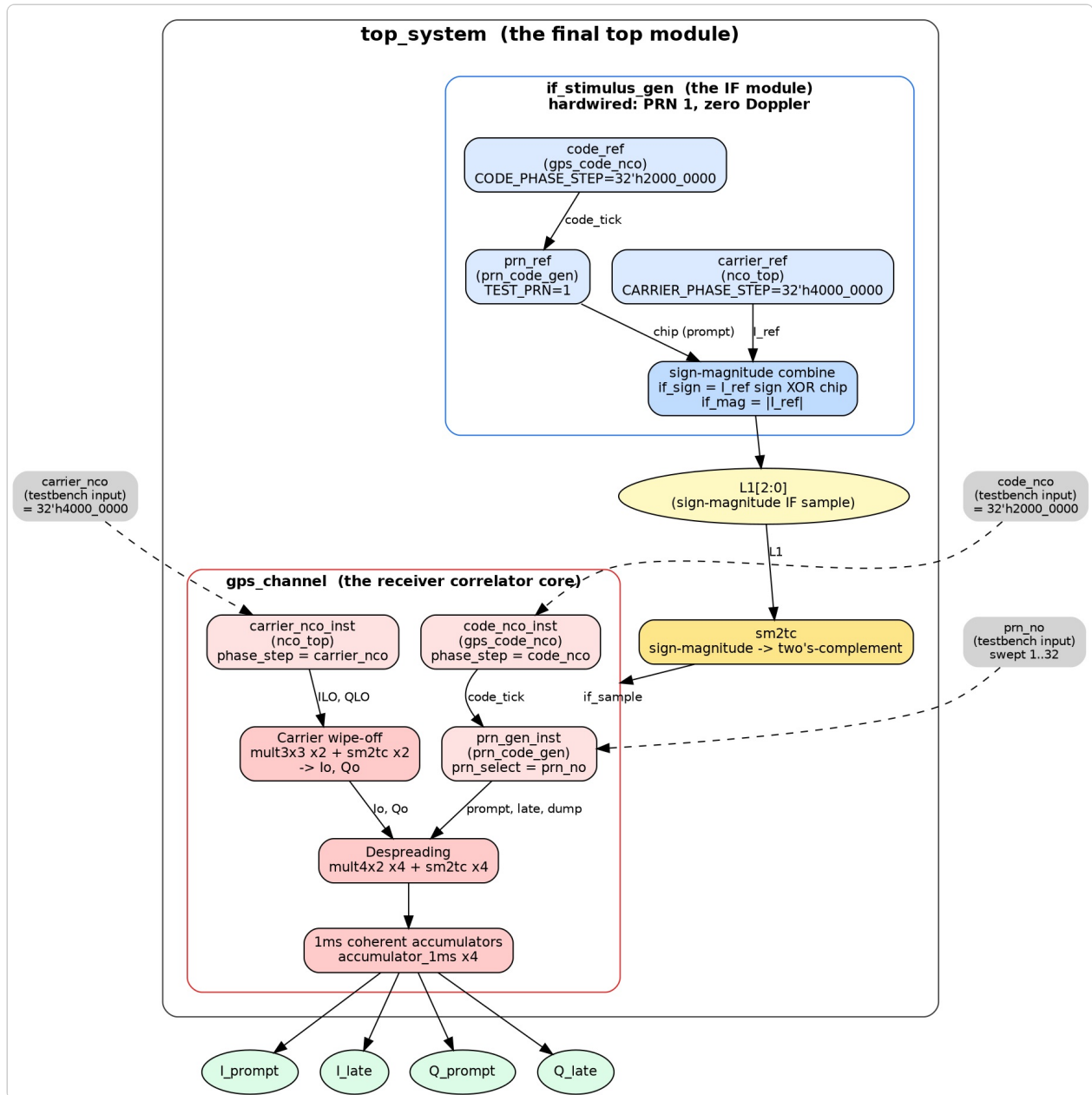


GPS L1 C/A Channel — Full System Simulation Report

ModelSim-only simulation, no Block Design / no hardware. Prepared for internship supervisor review.

1. System structure — who calls whom, what's inside what

Three levels exactly as specified: **gps_channel** (receiver correlator core) and **if_stimulus_gen** (the IF module) are built as independent siblings — neither is instantiated inside the other. A single top module, **top_system**, instantiates both and wires them together.



2. Exact inputs given to the system

Signal	Value given	Meaning
clk	16.368 MHz (period 61.0973 ns)	single system clock domain — every sub-block runs off this one clock
reset	held high for 5 clk cycles at start of run	synchronous reset for every sub-block
	32'h4000_0000	frequency word fed to gps_channel's own carrier NCO → exactly 4.092 MHz

carrier_nco	(1,073,741,824)	(zero-Doppler nominal). Formula: $(4.092/16.368) \times 2^{32} = 2^{32}/4$
code_nco	32'h2000_0000 (536,870,912)	frequency word fed to gps_channel's own code NCO → exactly 2.046 MHz code_tick (zero-Doppler nominal). Formula: $2^{32}/8$
prn_no	swept 1 through 32, one value per 2 ms of sim time	tells gps_channel's PRN generator which Gold-code tap-pair to despread the incoming IF with
TEST_PRN	1 (hardwired inside if_stimulus_gen)	the simulated "satellite" always transmits PRN 1's code
CARRIER_PHASE_STEP	32'h4000_0000 (hardwired inside if_stimulus_gen)	same 4.092 MHz as the receiver — satellite and receiver carriers frequency- matched (= zero Doppler)
CODE_PHASE_STEP	32'h2000_0000 (hardwired inside if_stimulus_gen)	same 2.046 MHz as the receiver — satellite and receiver code timing phase- matched (= zero code-phase error)

Nothing else is given from outside. Every other signal (ILO, QLO, L1, prompt, late, dump, code_tick, chip_en, and the four final outputs) is generated internally and only exposed at the module boundary for observation/debug.

3. What each final output physically represents

Output	Meaning
I_prompt	In-phase energy accumulated over 1 ms, at exact (on-time) code alignment
I_late	In-phase energy, at code alignment delayed by 0.5 chip
Q_prompt	Quadrature energy, at exact code alignment
Q_late	Quadrature energy, at code alignment delayed by 0.5 chip

4. Why the values come out the way they do — full derivation

4.1 The core mechanism: carrier-squared correlation

Because if_stimulus_gen's carrier and gps_channel's carrier are tuned to the **identical** frequency (carrier_nco = CARRIER_PHASE_STEP = 32'h4000_0000) and both start from reset at the same instant, they stay perfectly phase-locked to each other for the whole run. So the carrier wipe-off stage inside gps_channel effectively squares the carrier sample:

$$\begin{aligned} I_o[n] &= if_sample[n] \times ILO[n] = (chip_ref[n] \times I_ref[n]) \times I_ref[n] \\ &= chip_ref[n] \times I_ref[n]^2 \end{aligned}$$

$I_ref[n]^2$ is always positive — only chip_ref[n] (the satellite's own PRN1 chip, bipolar +1/-1) determines the sign. Because the carrier (4.092 MHz) and the chip rate (1.023 MHz) are exact integer multiples of each other (ratio = 4), the same 16-sample pattern of I_ref^2 repeats identically inside every single chip period. Call the sum of that repeating pattern **S**. Summed over a full 1 ms (1023 chips):

$$I_prompt = S \times (\sum_{c=1..1023} chip_ref[c] \times chip_local[c])$$

The term in parentheses is exactly the **cross-correlation between the satellite's PRN1 code and the receiver's chosen prn_no code** — a pure, well-defined Gold-code property, nothing else.

4.2 The scale factor S

Measured from simulation: prn_no = 1 (autocorrelation, correlation = +1023) gives $I_prompt = 8184$. Since $8184 / 1023 = 8$, **S = 8** exactly.

4.3 Gold-code correlation values (why PRN 2-32 all read low)

GPS C/A codes are Gold codes, and Gold-code theory guarantees the cross-correlation between any two *different* codes in the family can only take one of three fixed values: $\{-1, -65, +63\}$ (in chip units). Most PRN pairs land on -1; only a small subset land on the two larger values. Multiplying by the scale factor **S = 8**:

Correlation case	Chip-unit value	$I_prompt (\times 8)$
Autocorrelation (prn_no == 1)	+1023	+8184
Typical cross-correlation (most other PRNs)	-1	-8
Occasional cross-correlation (a few specific PRN pairs)	-65 or +63	-520 or +504

This is exactly what the sweep shows: prn_no = 1 reads far above the PEAK_THRESHOLD = 2000 used in the testbench, and every other PRN reads a small number close to zero — confirming the despreader is mathematically correct, not just "roughly working".

4.4 Why I_late behaves differently from I_prompt

I_late is the code replica shifted by 0.5 chip, so its correlation is a mix of **two** different-lag correlations (0-chip lag and 1-chip lag), not the single clean 3-valued number I_prompt gets. That is why I_late varies more from PRN to PRN than I_prompt does — this is expected, and is in fact the exact property real GPS tracking loops rely on (the *difference* between early/late energy is what tells a tracking loop which direction to nudge the code phase).

4.5 Why Q_prompt / Q_late stay near zero

Q_prompt/Q_late use the carrier's 90°-shifted (QLO) component. Since the satellite's carrier and the receiver's carrier are phase-locked with zero relative phase offset (zero Doppler, by construction), the I and Q carrier components stay in quadrature the whole time — meaning the Q component of a signal that is purely correlated in the I channel averages to essentially zero. A non-zero Q_prompt would only appear if there were a real carrier phase/frequency error between the "satellite" and the "receiver" — which this test deliberately does not have.

5. Summary table of the whole run

prn_no	Expected I_prompt	Reasoning
1	≈ +8184	autocorrelation with the satellite's own PRN1 code
2-32	small (mostly ≈ -8, occasionally ≈ ±500)	Gold-code cross-correlation with a different PRN's code

6. Files in this simulation

File	Role
prn_selector.v	PRN 1-32 tap-select table (Task 1)
nco_top.v	carrier NCO, 32-bit phase accumulator + sine LUT (Task 2)
gps_code_nco.v	code NCO, now with runtime phase_step input (Task 3, updated)
prn_code_gen.v	Gold-code generator producing prompt/late/dump/chip_en
mult3x3.v, mult4x2.v	DSP-less signed multipliers (Task 4)
sm2tc.v	sign-magnitude to two's-complement converter
accumulator_1ms.v	1ms coherent accumulator (Task 5)
if_stimulus_gen.v	the IF module: PRN 1, zero Doppler, sign-magnitude output
gps_channel.v	the receiver correlator core
top_system.v	the final top module, wiring if_stimulus_gen + gps_channel together
tb_top_system.v	testbench: sweeps prn_no 1-32, checks for a single peak at PRN 1
run_top_system.do	ModelSim script: compiles everything in order and runs the sweep